

SOUMIK ROY

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Technical Skills

Languages: Java, Python, JavaScript

Databases: MySQL

Libraries / Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, OpenCV, MediaPipe

Visualization: Power BI, Tableau, Matplotlib

Tools & Platforms: Git, GitHub, REST APIs

Experience

Web Developer Intern for Pratigya NGO

06/2023 – 07/2023

- Designed and developed responsive web content for Pratigya's official website during a 30-day remote internship from June 15, 2023, to July 15, 2023, significantly enhancing user interface design and website accessibility features.
- Successfully contributed to the NGO's educational mission by creating engaging digital solutions and improving online presence, directly supporting quality education initiatives for underprivileged children.

Web Developer Intern for EnerQuest Solutions

07/2025 - 10/2025

- Designed and implemented an AI chatbot using NLP and ML techniques with intent recognition, context management, and basic prompt engineering in Python using libraries like NLTK and scikit-learn.
- Optimized dialogue flow and query handling to improve response accuracy and efficiency, integrating simple AI logic with REST APIs for user interaction.

Projects

Divya-Drishti - AI-Powered Assistive Website for Accessibility | *Next.js, Twilio API, OpenCage API*

[Link](#)

- Designed and developed a data-driven AI assistive platform that supports visually impaired individuals through real-time obstacle detection and enables deaf-mute users to communicate via sign language recognition, leveraging computer vision and deep learning models.
- Implemented machine learning pipelines using YOLOv5 for object detection, built and trained sign language recognition models with curated datasets, and integrated geospatial APIs like OpenCage, MapBox along with Twilio for emergency intelligence and real-time decision support.

Hasth-Akshar - A Sign Language Detection Model for Devanagari | *MediaPipe, OpenCV, TensorFlow*

[Link](#)

- Developed a Hindi sign language detection system using stickman analysis, NLP, and deep learning to transcribe video gestures into Hindi text, bridging communication gaps for the hearing impaired.
- Utilized MediaPipe for pose estimation, OpenCV for computer vision processing, TensorFlow for model development, and NLP techniques with a custom video dataset. Applied robust preprocessing and feature extraction to enhance model reliability.

Achievements

- 1st Runner-Up — Zelestra x AWS ML Ascend Challenge 2025:** Placed in the top 50 out of 1,500+ teams by applying machine learning algorithms and building ensemble models.
- Top 100 — Amazon ML Challenge 2025:** Ranked among the top 100 nationwide by reducing the SMAPE value through custom model optimization.
- CodeClash Hackathon Finalist:** Secured a finalist position among 700+ teams by developing Divya-Drishti, an AI-powered assistive website for specially abled individuals.

Leadership

- Founder - Eureka Club, Vellore Institute of Technology:** Established a research and development club to foster innovation, leading 100+ members across multiple research initiatives.
- Core Technical Team Member - IoT Club, Vellore Institute of Technology:** Contributed to project development and built the club's official website as a key technical member.

Education

Bachelor of Technology in Computer Science and Engineering

08/2022 - 08/2026

Vellore Institute of Technology

CGPA: 8.91